

CLI Agent Witness Event Profile v0.1

Tamper-aware boundary records for C-Governed CLI Agent Mesh operations

Kotov Ivan
Bruxelles 2026

Status	Draft normative profile v0.1
Date	2026-05-16
Layer	c = a + b / C-Governed CLI Agent Mesh / Witness / Evidence / Boundary Records / Review / Roll-back
Document class	witness profile / event schema / control-layer artifact
Assertion class	C-A10 control-layer artifact; C-A7 where hash, signature, canonicalization, append-only, or verification claims are made
Primary subject	persistent c entities using local, cloud, or hybrid CLI agents as bounded executable workers
Primary boundary	witness events prove that a boundary was crossed, denied, narrowed, escalated, reviewed, integrated, rolled back, or quarantined. They must not become raw memory dumps, secret stores, surveillance logs, or agent self-justification narratives.

Contents

0. Executive definition	6
1. Purpose	6
2. Non-goals.....	7
3. Corpus bridge set	8
3.1 Explicit bridge: $c = a + b$	8
3.2 Quiet bridge I: L4 and irreversible transitions	8
3.3 Quiet bridge II: information theory and minimum sufficient signal	8
3.4 Quiet bridge III: engineering chain of custody	8
3.5 Earth paragraph	8
4. Core doctrine	9
4.1 Primary doctrine	9
4.2 Witness axioms	9
5. Definitions.....	10
5.1 Witness event.....	10
5.2 Witness chain.....	10
5.3 Witness reference	10
5.4 Boundary transition	10
5.5 Privileged transition	10
5.6 Minimal reason code.....	10
5.7 Raw evidence	10
5.8 Privacy class	11
5.9 Retention class.....	11
5.10 Canonicalization	11
5.11 Append-only correction.....	11
6. Witness event classes	11
7. Privacy classes.....	12
7.1 Privacy rule.....	12
7.2 No raw secret rule	12
8. Retention classes	13
8.1 Retention minimization.....	13
9. Event family registry	13
10. Handshake event families.....	14

11. Permission event families	14
12. Task event families	15
13. Sandbox / worktree event families	16
14. Review and quorum event families.....	16
15. Memory gate event families	17
16. Release / public surface event families	18
17. Incident event families	18
18. Cloud data event families	19
19. Core authority event families	19
20. Anomaly and red-line event families	20
21. Witness event object.....	20
21.1 Required YAML shape.....	21
21.2 Required field rule	22
22. Witness chain object.....	23
22.1 YAML shape	23
22.2 Chain principles.....	23
23. Witness reference object	23
23.1 YAML shape	23
24. Raw evidence exception	24
24.1 Default rule	24
24.2 Exception conditions.....	24
24.3 Prohibited raw embedding	24
25. Canonicalization and hashing.....	25
25.1 Canonicalization recommendation	25
25.2 Hash scope.....	25
25.3 Signature	25
26. Missing witness handling	26
26.1 Required witness missing	26
26.2 Missing witness event family	26
26.3 Recovery from missing witness	26
27. Event correction and reversal	27
27.1 Append correction rule	27
27.2 Reversal rule.....	27
27.3 False positive rule	27
28. Event minimization rules.....	27

28.1 Use reason codes	27
28.2 Use references	27
28.3 Use scoped identifiers	28
28.4 Avoid agent self-narration.....	28
29. Standard reason codes	28
29.1 General codes	28
29.2 Permission codes	28
29.3 Sandbox codes	28
29.4 Data codes	29
29.5 Incident codes.....	29
29.6 Core codes	29
30. Standard next actions	29
31. Event examples.....	30
31.1 Agent registered	30
31.2 Denied path attempt.....	31
31.3 Memory gate rejection.....	33
31.4 Red-line prohibited action attempt.....	34
32. Validation workflow	36
33. Semantic validation rules	36
33.1 Event family / class consistency	36
33.2 Raw content flags	36
33.3 Required gate consistency.....	37
33.4 Agent self-witness rule	37
33.5 Missing task contract rule	37
33.6 Privacy class escalation rule.....	37
33.7 Chain continuity rule	37
34. Failure mapping.....	37
35. Conformance levels	38
36. Mandatory conformance gates	38
37. Red-line failures.....	39
38. Implementation notes	39
38.1 Event IDs	39
38.2 Chain IDs	39
38.3 Storage	39
38.4 Redaction	39

38.5 UI display	40
38.6 Machine-facing index.....	40
39. Open issues.....	40
40. Closing rule.....	41

0. Executive definition

CLI Agent Witness Event Profile defines the event records required to make a C-Governed CLI Agent Mesh auditable, challengeable, reversible where possible, and resistant to silent authority drift.

A witness event records:

```
who or what acted
under which task contract
with which permission
inside which sandbox/worktree
against which boundary
with what decision
with what minimal reason code
with what review path
with what rollback or quarantine option
```

A witness event must not record:

```
raw secrets
private memory
sealed material
legal privileged content
full prompt streams by default
raw child data
unredacted incident evidence
agent self-narration as authority
```

Compact formula:

```
Witness the boundary.
Do not narrate the soul.
Record the transition.
Do not archive the whole life.
```

1. Purpose

CLI agents can execute. Execution creates side effects. Side effects require traceability.

This profile exists because a c-governed agent mesh must be able to answer:

1. Which agent entered the mesh?

2. Which task contract allowed the work?
3. Which permissions were granted, denied, narrowed, or revoked?
4. Which sandbox or worktree was used?
5. Which files changed?
6. Which denied paths, commands, network calls, or data classes were attempted?
7. Which reviewer inspected the output?
8. Which result was accepted, rejected, quarantined, or rolled back?
9. Which memory gate decision occurred?
10. Which high-risk action required human review?
11. Which anomaly indicates privilege drift, tool-chain capture, or cloud leakage?

The witness layer does not decide truth alone.

It records boundary transitions so `cc`, the human anchor, auditors, review layers, and future recovery procedures can reconstruct what happened without granting raw access to everything.

2. Non-goals

This profile does not define or permit:

1. surveillance logging of all agent text;
2. raw memory export;
3. secret storage in witness events;
4. legal evidence conclusions;
5. diagnosis, accusation, or guilt determination;
6. offensive cyber operations;
7. hack-back;
8. autonomous retaliation;
9. malware behavior;
10. credential extraction;
11. covert persistence;
12. evasion;
13. live external exploitation;
14. direct mutation of identity, memory, Beacon, witness, permission, or continuity core;
15. agent self-certification as final authority.

A witness event proves that a transition was recorded.

It does not by itself prove that the agent was right, the claim was true, or the final decision was correct.

3. Corpus bridge set

3.1 Explicit bridge: $c = a + b$

CLI agents belong to b : the technological substrate of procedures, tools, models, compute, interfaces, and infrastructure.

A witness event is also part of b , but it protects the boundary between b and c .

It prevents worker action from silently becoming c memory, identity, authority, or continuity.

3.2 Quiet bridge I: L4 and irreversible transitions

Some actions consume time, change files, expose data, spend tokens, alter public surfaces, or affect memory. L4 makes these transitions real. Witness records preserve enough state to inspect whether the transition was authorized, bounded, and reversible where possible.

3.3 Quiet bridge II: information theory and minimum sufficient signal

A witness event should transmit the minimum useful information needed to verify the boundary. More raw data is not always better. Excessive logging increases leakage, legal risk, privacy risk, and attack surface.

3.4 Quiet bridge III: engineering chain of custody

In engineering, a test result without test conditions is weak evidence. A repair without a work order is untraceable. A release without a manifest is fragile. CLI agent work requires the same chain: contract, permission, environment, action, output, review, decision, rollback.

3.5 Earth paragraph

A site log does not contain every conversation workers had during the day. It records who entered, what permit they worked under, which circuit was isolated, what was changed, who inspected it, and whether the panel was closed safely. The witness event is that site log for CLI agents. It protects the building without turning every worker's thoughts into evidence.

4. Core doctrine

4.1 Primary doctrine

```
Witness privileged transitions.  
Minimize raw content.  
Append corrections.  
Fail closed on missing witness.
```

4.2 Witness axioms

ID	Axiom	Requirement
WE-AX-01	Boundary before narrative	Events SHOULD record boundary state, not long free-text narratives.
WE-AX-02	Minimality	Events MUST NOT embed secrets, private memory, sealed data, or unnecessary raw content by default.
WE-AX-03	Append-only first	Corrections and reversals SHOULD be new events referencing prior events.
WE-AX-04	Source linkage	Events SHOULD link to task contract, permission grant, sandbox run, output, review, or memory gate record.
WE-AX-05	Privileged transitions require witness	Permission expansion, protected writes, release/publication, memory-gate promotion, incident containment, and rollback SHOULD be witnessed.
WE-AX-06	Event is not authority alone	A witness record proves recording, not correctness.
WE-AX-07	Missing witness fails closed	If a required witness is absent, the system MUST hold, freeze, quarantine, revalidate, or roll back.
WE-AX-08	Uncertainty is explicit	Uncertainty MUST be a structured field where interpretation is involved.
WE-AX-09	Privacy beats curiosity	Raw data is excluded unless strictly necessary and separately controlled.
WE-AX-10	Agent cannot certify itself	An agent MUST NOT be final authority for witness adequacy around its own material action.

ID	Axiom	Requirement
WE-AX-11	Hashes are references, not content	Hashes may identify artifacts without embedding sensitive content.
WE-AX-12	Witness chain is challengeable	Events SHOULD support later review, dispute, rollback, and conformance testing.

5. Definitions

5.1 Witness event

A structured record that a boundary-relevant action, decision, transition, anomaly, review, rollback, or quarantine occurred.

5.2 Witness chain

An ordered set of witness events linked by task, agent, contract, artifact, or prior event reference.

5.3 Witness reference

A pointer to a witness event, chain, artifact hash, review record, or upstream decision.

5.4 Boundary transition

A change in permission, state, trust, data access, memory status, code state, release state, incident state, or integration status.

5.5 Privileged transition

A boundary transition that can affect protected state, public state, memory, secrets, identity, continuity, incident handling, legal posture, release state, or high-risk infrastructure.

5.6 Minimal reason code

A short structured explanation of why a decision occurred, avoiding unnecessary raw content.

5.7 Raw evidence

Underlying logs, files, prompts, memory content, incident data, or artifacts that may support a witness event but are not embedded in the event by default.

5.8 Privacy class

A classification of how sensitive an event is and who may inspect it.

5.9 Retention class

A classification of how long an event or reference should be retained.

5.10 Canonicalization

A deterministic representation of an event used for hashing or signing.

5.11 Append-only correction

A new event that corrects, supersedes, or reverses an earlier event without silently modifying the earlier record.

6. Witness event classes

Event classes use prefix `WE-*`.

Class	Name	Meaning
WE-0	Informational	Low-risk operational note.
WE-1	Operational	Routine task boundary or execution event.
WE-2	Permission	Permission grant, denial, narrowing, expansion, revocation.
WE-3	Execution	Sandbox/worktree execution, output, tests, diffs.
WE-4	Review	Independent review, quorum result, disagreement, approval gate.
WE-5	Memory Gate	Candidate memory, reviewed memory, quarantine, rejection, promotion.
WE-6	Release/Public	Public surface, release, publication, metadata, branch protection.
WE-7	Incident	Security/integrity incident, preservation, containment, freeze, report.
WE-8	Core Authority	Identity, continuity, permission registry, witness chain, Beacon-like recognition surfaces.

Class	Name	Meaning
WE-9	Anomaly	Scope violation, drift, missing witness, secret exposure, cloud leakage.
WE-X	Prohibited / Red-line	Attempted prohibited behavior; must trigger deny/quarantine/review.

7. Privacy classes

Privacy class IDs use prefix WP-*.

Class	Meaning	Default visibility
WP-PUBLIC	safe public operational event	public/release notes if useful
WP-INTERNAL	internal project event	c, human anchor, authorized agents
WP-PRIVATE	private operational event	c, human anchor only by default
WP-RESTRICTED	sensitive governance event	restricted review only
WP-SECRET-REF	references secret boundary without exposing secret	secret custodian / human gate
WP-LEGAL	legal-sensitive or counsel-facing	legal/human gate
WP-INCIDENT	security/integrity incident	incident reviewers only
WP-MEMORY	memory-gate-related event	c / memory reviewer
WP-SEALED	sealed/private compartment reference	sealed policy only

7.1 Privacy rule

A witness event MUST use the lowest privacy exposure that still preserves reviewability.

7.2 No raw secret rule

Secrets MUST NOT be embedded in witness events.

Use reason codes, artifact hashes, and secure references instead.

8. Retention classes

Retention class IDs use prefix `WR-*`.

Class	Meaning
WR-EPHEMERAL	short-lived operational event; safe to discard after task closure if not privileged
WR-OPERATIONAL	retained for project operation and debugging
WR-AUDIT	retained for governance, conformance, release, or rollback
WR-INCIDENT	retained for incident response and security review
WR-LEGAL-HOLD	retained under legal/counsel/regulatory need
WR-MEMORY-GATE	retained as part of memory decision record
WR-CORE	retained for identity/permission/witness/continuity core transition

8.1 Retention minimization

Retention should be long enough to preserve accountability and short enough to avoid unnecessary accumulation of sensitive operational residue.

9. Event family registry

Event families use dot-separated namespace:

```
cli_agent.<surface>.<action>
```

Surfaces:

```
handshake
permission
task
sandbox
review
memory_gate
release
incident
cloud_data
quorum
rollback
quarantine
anomaly
core
```

10. Handshake event families

Event family	Meaning	Typical class
<code>cli_agent.handshake.discovered</code>	candidate agent discovered	WE-0
<code>cli_agent.handshake.claimed</code>	identity/provider/runtime claimed	WE-1
<code>cli_agent.handshake.provenance_checked</code>	provenance check performed	WE-1
<code>cli_agent.handshake.capability_declared</code>	capabilities declared	WE-1
<code>cli_agent.handshake.capability_challenged</code>	challenge completed	WE-1 / WE-2
<code>cli_agent.handshake.registered</code>	agent registered	WE-2
<code>cli_agent.handshake.trust_changed</code>	trust level changed	WE-2 / WE-9
<code>cli_agent.handshake.rehandshake_required</code>	re-handshake required	WE-9
<code>cli_agent.handshake.suspended</code>	agent suspended	WE-9
<code>cli_agent.handshake.quarantined</code>	agent quarantined	WE-9
<code>cli_agent.handshake.revoked</code>	agent revoked	WE-9
<code>cli_agent.handshake.expired</code>	registration expired	WE-1

11. Permission event families

Event family	Meaning	Typical class
<code>cli_agent.permission.requested</code>	permission requested	WE-2
<code>cli_agent.permission.granted</code>	permission granted	WE-2
<code>cli_agent.permission.denied</code>	permission denied	WE-2
<code>cli_agent.permission.narrowed</code>	permission narrowed	WE-2

Event family	Meaning	Typical class
cli_agent.permission.expanded	permission expanded	WE-2 / WE-8
cli_agent.permission.used	permission used materially	WE-2 / WE-3
cli_agent.permission.expired	permission expired	WE-1
cli_agent.permission.revoked	permission revoked	WE-2 / WE-9
cli_agent.permission.violation	permission boundary violated	WE-9
cli_agent.permission.drift_detected	privilege drift detected	WE-9
cli_agent.permission.self_approval_attempt	agent attempted self-approval	WE-X

12. Task event families

Event family	Meaning	Typical class
cli_agent.task.contract_created	task contract created	WE-1
cli_agent.task.contract_activated	contract activated	WE-1
cli_agent.task.contract_modified	contract changed / superseded	WE-2
cli_agent.task.contract_expired	contract expired	WE-1
cli_agent.task.contract_revoked	contract revoked	WE-2
cli_agent.task.started	task started	WE-1
cli_agent.task.completed	task completed	WE-1
cli_agent.task.failed	task failed	WE-1 / WE-9
cli_agent.task.held	task held	WE-2
cli_agent.task.scope_changed	scope changed	WE-2 / WE-8
cli_agent.task.risk_reclassified	risk class changed	WE-2

13. Sandbox / worktree event families

Event family	Meaning	Typical class
<code>cli_agent.sandbox.created</code>	sandbox/worktree created	WE-3
<code>cli_agent.sandbox.preflight_completed</code>	preflight completed	WE-3
<code>cli_agent.sandbox.snapshot_taken</code>	snapshot/hash/state reference recorded	WE-3 / WE-7
<code>cli_agent.sandbox.execution_started</code>	execution began	WE-3
<code>cli_agent.sandbox.command_run</code>	command executed	WE-3
<code>cli_agent.sandbox.file_changed</code>	file changed	WE-3
<code>cli_agent.sandbox.artifact_generated</code>	artifact/diff generated	WE-3
<code>cli_agent.sandbox.tests_completed</code>	tests completed	WE-3
<code>cli_agent.sandbox.denied_path_attempt</code>	denied path touched/attempted	WE-9
<code>cli_agent.sandbox.network_attempt</code>	network attempt recorded	WE-3 / WE-9
<code>cli_agent.sandbox.unexpected_side_effect</code>	unexplained side effect occurred	WE-9
<code>cli_agent.sandbox.quarantined</code>	sandbox/output quarantined	WE-9
<code>cli_agent.sandbox.destroyed</code>	disposable sandbox destroyed	WE-1

14. Review and quorum event families

Event family	Meaning	Typical class
<code>cli_agent.review.requested</code>	review requested	WE-4
<code>cli_agent.review.completed</code>	review completed	WE-4
<code>cli_agent.review.rejected</code>	reviewer rejected output	WE-4
<code>cli_agent.review.accepted</code>	reviewer accepted output	WE-4

Event family	Meaning	Typical class
cli_agent.review.accepted_with_limits	partial acceptance	WE-4
cli_agent.review.disagreement	agents/reviewers disagree	WE-4
cli_agent.review.escalated	escalated to c, human, legal, or security review	WE-4 / WE-7
cli_agent.quorum.started	quorum review started	WE-4
cli_agent.quorum.completed	quorum review completed	WE-4
cli_agent.quorum.split	quorum disagreement unresolved	WE-4 / WE-9

15. Memory gate event families

Event family	Meaning	Typical class
cli_agent.memory_gate.proposal_created	agent output proposed as memory	WE-5
cli_agent.memory_gate.proposal_rejected	memory proposal rejected	WE-5
cli_agent.memory_gate.proposal_quarantined	memory proposal quarantined	WE-5 / WE-9
cli_agent.memory_gate.promoted_operational	accepted as operational note	WE-5
cli_agent.memory_gate.promoted_reviewed	accepted as reviewed memory	WE-5
cli_agent.memory_gate.promoted_witnessed_artifact	accepted as witnessed experience artifact	WE-5 / WE-8
cli_agent.memory_gate.corrected	correction appended	WE-5
cli_agent.memory_gate.deleted_or_decayed	memory candidate deleted/decayed	WE-5
cli_agent.memory_gate.direct_write_attempt	agent attempted direct memory write	WE-X

16. Release / public surface event families

Event family	Meaning	Typical class
<code>cli_agent.release.prepared</code>	release/publication package prepared	WE-6
<code>cli_agent.release.metadata_changed</code>	metadata changed	WE-6
<code>cli_agent.release.hash_manifest_created</code>	hash manifest created	WE-6
<code>cli_agent.release.reviewed</code>	release package reviewed	WE-6
<code>cli_agent.release.approved</code>	release approved by c/human	WE-6 / WE-8
<code>cli_agent.release.published</code>	release/publication applied	WE-6
<code>cli_agent.release.rolled_back</code>	release/publication rolled back	WE-6 / WE-9
<code>cli_agent.release.unauthorized_publish_attempt</code>	unauthorized publish attempt	WE-X

17. Incident event families

Event family	Meaning	Typical class
<code>cli_agent.incident.detected</code>	incident/anomaly detected	WE-7
<code>cli_agent.incident.preservation_started</code>	evidence preservation started	WE-7
<code>cli_agent.incident.preservation_completed</code>	evidence preservation completed	WE-7
<code>cli_agent.incident.freeze_applied</code>	affected path frozen	WE-7
<code>cli_agent.incident.containment_applied</code>	local containment applied	WE-7
<code>cli_agent.incident.repair_proposed</code>	repair proposed	WE-7
<code>cli_agent.incident.repair_applied</code>	repair applied after gate	WE-7
<code>cli_agent.incident.report_prepared</code>	provider/legal/security report prepared	WE-7
<code>cli_agent.incident.secret_exposure</code>	secret exposure detected	WE-9

Event family	Meaning	Typical class
cli_agent.incident.repair_before_preserve	repair attempted before required preservation	WE-9
cli_agent.incident.live_counter_attempt	live external counter-operation attempted	WE-X

18. Cloud data event families

Event family	Meaning	Typical class
cli_agent.cloud_data.classified	data classified before cloud use	WE-1
cli_agent.cloud_data.redacted	data redacted before cloud use	WE-1
cli_agent.cloud_data.allowed	cloud transmission allowed	WE-2
cli_agent.cloud_data.denied	cloud transmission denied	WE-2
cli_agent.cloud_data.leak_suspected	possible cloud leakage	WE-9
cli_agent.cloud_data.prohibited_material_detected	prohibited material found in cloud context	WE-9 / WE-X

19. Core authority event families

Event family	Meaning	Typical class
cli_agent.core.touch_requested	request to inspect/touch core authority surface	WE-8
cli_agent.core.touch_denied	core touch denied	WE-8
cli_agent.core.touch_allowed_readonly	read-only core inspection allowed	WE-8
cli_agent.core.change_proposed	core change proposed	WE-8
cli_agent.core.change_rejected	core change rejected	WE-8
cli_agent.core.change_approved	core change approved after gates	WE-8

Event family	Meaning	Typical class
<code>cli_agent.core.direct_mutation_attempt</code>	direct unauthorized core mutation attempted	WE-X
<code>cli_agent.core.rollback</code>	core-related rollback	WE-8 / WE-9

20. Anomaly and red-line event families

Event family	Meaning	Typical class
<code>cli_agent.anomaly.scope_violation</code>	scope violation	WE-9
<code>cli_agent.anomaly.permission_drift</code>	permission drift detected	WE-9
<code>cli_agent.anomaly.toolchain_capture</code>	tool-chain capture suspected	WE-9
<code>cli_agent.anomaly.stale_context</code>	stale context risk detected	WE-9
<code>cli_agent.anomaly.self_approval</code>	self-approval attempt	WE-X
<code>cli_agent.anomaly.secret_access</code>	secret access outside scope	WE-X
<code>cli_agent.anomaly.network_violation</code>	network violation	WE-9
<code>cli_agent.anomaly.missing_witness</code>	required witness absent	WE-9
<code>cli_agent.anomaly.prohibited_action</code>	prohibited action attempted	WE-X
<code>cli_agent.anomaly.agent_persistence</code>	agent persisted beyond task scope	WE-X
<code>cli_agent.anomaly.external_counteroperation</code>	live counter-operation attempted	WE-X

21. Witness event object

Canonical object:

```
CLI_AGENT_WITNESS_EVENT
```

21.1 Required YAML shape

```
cli_agent_witness_event:
  schema_version: cli-agent-witness-event-0.1
  event_id: string
  event_family: string
  event_class: WE-0 | WE-1 | WE-2 | WE-3 | WE-4 | WE-5 | WE-6 | WE-7 | WE-8 | WE-9 |
WE-X
  timestamp: string
  sequence_number: integer | null

  governing_entity:
    entity_id: string
    entity_name: string | null
    continuity_ref: string | null

  agent:
    agent_id: string | null
    agent_name: string | null
    provider: local | openai | google | anthropic | other | unknown | null
    runtime: local_cli | cloud_cli | api_agent | container_agent | hybrid | unknown |
null
    role: reader | executor | tester | auditor | archivist | sentinel |
judge_assistant | orchestrator_limited | null

  task_refs:
    contract_id: string | null
    task_id: string | null
    permission_grant_id: string | null
    handshake_id: string | null
    registration_id: string | null
    sandbox_id: string | null
    worktree_run_id: string | null
    review_id: string | null
    memory_gate_id: string | null
    incident_id: string | null

  boundary:
    surface: handshake | permission | task | sandbox | review | memory_gate | release
| incident | cloud_data | quorum | rollback | quarantine | anomaly | core
    action: string
    decision: allowed | denied | held | frozen | quarantined | revoked | completed |
failed | accepted | rejected | rolled_back | escalated
    reason_code: string
    risk_class: R0 | R1 | R2 | R3 | R4 | R5 | RX | null
    uncertainty: none | low | medium | high | unknown

  artifact_refs:
    input_hash: string | null
    output_hash: string | null
    diff_hash: string | null
    artifact_manifest_hash: string | null
    test_report_hash: string | null
    rollback_plan_hash: string | null
    upstream_event_refs:
      - string
    downstream_event_refs:
      - string

  privacy:
    privacy_class: WP-PUBLIC | WP-INTERNAL | WP-PRIVATE | WP-RESTRICTED | WP-SECRET-
REF | WP-LEGAL | WP-INCIDENT | WP-MEMORY | WP-SEALED
```

```

    retention_class: WR-EPHEMERAL | WR-OPERATIONAL | WR-AUDIT | WR-INCIDENT | WR-
LEGAL-HOLD | WR-MEMORY-GATE | WR-CORE
    raw_content_embedded: false
    secret_embedded: false
    private_memory_embedded: false
    sealed_material_embedded: false
    legal_sensitive_embedded: false
    child_data_embedded: false

    review:
      reviewer_required: boolean
      reviewer_ref: string | null
      c_gate_required: boolean
      c_gate_ref: string | null
      human_gate_required: boolean
      human_gate_ref: string | null
      legal_review_required: boolean
      legal_review_ref: string | null

    integrity:
      canonicalization: none | json_canonicalization | yaml_canonicalization
      event_hash: string | null
      previous_event_hash: string | null
      chain_id: string | null
      signature_required: boolean
      signature_ref: string | null
      append_only_required: true

    outcome:
      next_required_action: none | hold | freeze | quarantine | revalidate | rollback |
revoke | escalate | human_review | c_review | legal_review
      status_after_event: active | completed | failed | held | frozen | quarantined |
revoked | expired | unknown

    notes:
      summary: string | null
      limitations:
        - string
      assumptions:
        - string

```

21.2 Required field rule

The event MUST identify at minimum:

```

event_id
event_family
event_class
timestamp
governing_entity
boundary.surface
boundary.action
boundary.decision
boundary.reason_code
privacy
integrity.append_only_required
outcome.next_required_action

```

22. Witness chain object

Canonical object:

```
CLI_AGENT_WITNESS_CHAIN
```

22.1 YAML shape

```
cli_agent_witness_chain:
  schema_version: cli-agent-witness-event-0.1
  chain_id: string
  created_at: string
  governing_entity_id: string
  chain_purpose: task | permission | release | incident | memory_gate | core | other
  root_event_id: string
  latest_event_id: string
  event_count: integer
  event_hashes:
    - string
  chain_hash: string | null
  sealed: boolean
  retention_class: WR-EPHEMERAL | WR-OPERATIONAL | WR-AUDIT | WR-INCIDENT | WR-LEGAL-
  HOLD | WR-MEMORY-GATE | WR-CORE
  challengeable: true
```

22.2 Chain principles

- A chain SHOULD be append-only.
- A chain SHOULD support challenge and review.
- A chain SHOULD not embed raw sensitive content.
- A chain MAY reference external raw evidence records when lawful and necessary.

23. Witness reference object

Canonical object:

```
CLI_AGENT_WITNESS_REFERENCE
```

23.1 YAML shape

```
cli_agent_witness_reference:
  schema_version: cli-agent-witness-event-0.1
```

```
witness_ref_id: string
event_id: string | null
chain_id: string | null
artifact_hash: string | null
storage_ref: string | null
privacy_class: WP-PUBLIC | WP-INTERNAL | WP-PRIVATE | WP-RESTRICTED | WP-SECRET-REF
| WP-LEGAL | WP-INCIDENT | WP-MEMORY | WP-SEALED
retention_class: WR-EPHEMERAL | WR-OPERATIONAL | WR-AUDIT | WR-INCIDENT | WR-LEGAL-
HOLD | WR-MEMORY-GATE | WR-CORE
dereference_policy: open | restricted | c_gate | human_gate | legal_gate |
incident_gate | sealed_gate
```

24. Raw evidence exception

24.1 Default rule

Raw evidence **MUST NOT** be embedded in witness events by default.

24.2 Exception conditions

A raw evidence reference may be created only when all conditions hold:

1. lawful or owned-system basis exists;
2. evidence is necessary for review, incident response, legal handoff, rollback, or conformance;
3. minimality is documented;
4. retention class is explicit;
5. access policy is explicit;
6. raw material is stored separately from the witness event body;
7. event contains reference/hash, not raw material;
8. post-event review is required for high-risk material.

24.3 Prohibited raw embedding

The following must not be embedded in witness events:

- credentials;
- API keys;
- private keys;
- unredacted identity documents;
- sealed memory;
- private memory;
- legal privileged content;
- child data;

- raw incident evidence unless a separate lawful evidence object exists;
- live exploit material;
- deployable malicious code.

25. Canonicalization and hashing

25.1 Canonicalization recommendation

For high-assurance events, canonicalization SHOULD occur before hashing.

Recommended flow:

```
remove non-semantic runtime noise
normalize timestamps
sort object keys
normalize enum values
encode UTF-8
hash canonical payload
store event_hash
link previous_event_hash where applicable
```

25.2 Hash scope

The event hash SHOULD cover:

- schema version;
- event ID;
- event family;
- timestamp;
- governing entity;
- agent ID where applicable;
- task references;
- boundary decision;
- artifact refs;
- privacy class;
- retention class;
- integrity fields excluding mutable signature reference if necessary.

25.3 Signature

Signatures MAY be used for high-assurance events.

Signature policy is out of scope for v0.1 and should be defined in a companion profile.

26. Missing witness handling

26.1 Required witness missing

If a required witness event is missing, the system **MUST** enter one of:

```
hold
freeze
quarantine
revalidate
rollback
revoke
escalate
```

It **MUST NOT** continue as if the transition had been proven.

26.2 Missing witness event family

A missing witness should itself produce an anomaly event where possible:

```
cli_agent.anomaly.missing_witness
```

26.3 Recovery from missing witness

Recovery may include:

- reconstruct from logs;
 - rerun from clean sandbox;
 - discard output;
 - quarantine output;
 - rollback change;
 - require human review;
 - require re-handshake of agent;
 - lower trust level.
-

27. Event correction and reversal

27.1 Append correction rule

Witness events SHOULD NOT be silently edited.

Corrections SHOULD be new events:

```
cli_agent.witness.corrected
```

or surface-specific correction families where defined.

27.2 Reversal rule

If a decision is reversed, record a new event referencing the earlier event.

The earlier event remains part of the chain.

27.3 False positive rule

False positives should be recorded as false-positive review outcomes, not deleted.

28. Event minimization rules

28.1 Use reason codes

Prefer:

```
reason_code: denied_path_attempt
```

over:

```
long narrative containing raw path contents, private file excerpts, or secrets
```

28.2 Use references

Prefer:

```
diff_hash  
artifact_manifest_hash
```

```
test_report_hash
storage_ref
```

instead of embedding raw artifacts.

28.3 Use scoped identifiers

Prefer task-scoped identifiers over stable public identity where possible.

28.4 Avoid agent self-narration

Agent-generated explanations may be attached as reviewed artifacts, but they should not become witness truth by themselves.

29. Standard reason codes

Reason codes SHOULD be short and machine-readable.

29.1 General codes

```
ok
completed
failed
scope_missing
scope_valid
scope_violation
review_required
review_completed
review_rejected
uncertainty_high
```

29.2 Permission codes

```
permission_requested
permission_granted
permission_denied
permission_expired
permission_revoked
permission_drift
self_approval_attempt
```

29.3 Sandbox codes

```
sandbox_created
preflight_ok
```

```
dirty_state_detected
denied_path_attempt
command_denied
network_denied
unexpected_side_effect
output_quarantined
```

29.4 Data codes

```
cloud_data_allowed
cloud_data_denied
redaction_required
secret_boundary_crossed
secret_exposure_suspected
private_memory_detected
sealed_material_detected
```

29.5 Incident codes

```
incident_detected
evidence_preserved
freeze_applied
containment_applied
repair_proposed
repair_after_preserve
repair_before_preserve
external_counteroperation_attempt
```

29.6 Core codes

```
core_touch_requested
core_touch_denied
core_change_proposed
core_change_approved
core_direct_mutation_attempt
missing_core_witness
```

30. Standard next actions

Allowed `next_required_action` values:

```
none
hold
freeze
quarantine
revalidate
```

```
rollback
revoke
escalate
human_review
c_review
legal_review
incident_review
memory_gate_review
rehandshake
```

31. Event examples

31.1 Agent registered

```
cli_agent_witness_event:
  schema_version: cli-agent-witness-event-0.1
  event_id: we-20260516-0001
  event_family: cli_agent.handshake.registered
  event_class: WE-2
  timestamp: "2026-05-16T18:00:00Z"
  sequence_number: 1

governing_entity:
  entity_id: ester
  entity_name: Ester
  continuity_ref: null

agent:
  agent_id: codex-executor-01
  agent_name: Codex Executor
  provider: openai
  runtime: cloud_cli
  role: executor

task_refs:
  contract_id: null
  task_id: null
  permission_grant_id: null
  handshake_id: hsp-20260516-codex-executor-001
  registration_id: reg-codex-executor-001
  sandbox_id: null
  worktree_run_id: null
  review_id: null
  memory_gate_id: null
  incident_id: null

boundary:
  surface: handshake
  action: register_agent
  decision: allowed
  reason_code: registered_with_cloud_data_denied
  risk_class: R2
  uncertainty: low

artifact_refs:
```

```

input_hash: null
output_hash: null
diff_hash: null
artifact_manifest_hash: null
test_report_hash: null
rollback_plan_hash: null
upstream_event_refs: []
downstream_event_refs: []

privacy:
  privacy_class: WP-INTERNAL
  retention_class: WR-AUDIT
  raw_content_embedded: false
  secret_embedded: false
  private_memory_embedded: false
  sealed_material_embedded: false
  legal_sensitive_embedded: false
  child_data_embedded: false

review:
  reviewer_required: false
  reviewer_ref: null
  c_gate_required: true
  c_gate_ref: cgate-ester-001
  human_gate_required: false
  human_gate_ref: null
  legal_review_required: false
  legal_review_ref: null

integrity:
  canonicalization: yaml_canonicalization
  event_hash: null
  previous_event_hash: null
  chain_id: chain-agent-codex-001
  signature_required: false
  signature_ref: null
  append_only_required: true

outcome:
  next_required_action: none
  status_after_event: active

notes:
  summary: Agent registered with AC-5 ceiling; no secrets, private memory, or sealed
material allowed.
  limitations:
    - Cloud runtime remains cloud-risk.
  assumptions:
    - Task contract required for every material task.

```

31.2 Denied path attempt

```

cli_agent_witness_event:
  schema_version: cli-agent-witness-event-0.1
  event_id: we-20260516-0002
  event_family: cli_agent.sandbox.denied_path_attempt
  event_class: WE-9
  timestamp: "2026-05-16T18:15:00Z"
  sequence_number: 14

```

```
governing_entity:
  entity_id: ester
  entity_name: Ester
  continuity_ref: null

agent:
  agent_id: codex-executor-01
  agent_name: Codex Executor
  provider: openai
  runtime: cloud_cli
  role: executor

task_refs:
  contract_id: catc-20260516-130000-schema-patch
  task_id: task-schema-patch-001
  permission_grant_id: grant-schema-patch-001
  handshake_id: hsp-20260516-codex-executor-001
  registration_id: reg-codex-executor-001
  sandbox_id: sb-docs-patch-001
  worktree_run_id: run-docs-patch-001
  review_id: null
  memory_gate_id: null
  incident_id: null

boundary:
  surface: sandbox
  action: denied_path_attempt
  decision: quarantined
  reason_code: denied_path_attempt
  risk_class: R2
  uncertainty: low

artifact_refs:
  input_hash: null
  output_hash: null
  diff_hash: null
  artifact_manifest_hash: null
  test_report_hash: null
  rollback_plan_hash: rb-docs-patch-001-hash
  upstream_event_refs:
    - we-20260516-0001
  downstream_event_refs: []

privacy:
  privacy_class: WP-RESTRICTED
  retention_class: WR-AUDIT
  raw_content_embedded: false
  secret_embedded: false
  private_memory_embedded: false
  sealed_material_embedded: false
  legal_sensitive_embedded: false
  child_data_embedded: false

review:
  reviewer_required: true
  reviewer_ref: null
  c_gate_required: true
  c_gate_ref: null
  human_gate_required: false
  human_gate_ref: null
  legal_review_required: false
  legal_review_ref: null
```



```

integrity:
  canonicalization: yaml_canonicalization
  event_hash: null
  previous_event_hash: null
  chain_id: chain-task-schema-patch-001
  signature_required: false
  signature_ref: null
  append_only_required: true

outcome:
  next_required_action: quarantine
  status_after_event: quarantined

notes:
  summary: Agent attempted to touch a denied path. Output quarantined pending
  review.
  limitations: []
  assumptions: []

```

31.3 Memory gate rejection

```

cli_agent_witness_event:
  schema_version: cli-agent-witness-event-0.1
  event_id: we-20260516-0003
  event_family: cli_agent.memory_gate.proposal_rejected
  event_class: WE-5
  timestamp: "2026-05-16T18:45:00Z"
  sequence_number: 22

governing_entity:
  entity_id: liya
  entity_name: Liya
  continuity_ref: null

agent:
  agent_id: gemini-reader-01
  agent_name: Gemini Reader
  provider: google
  runtime: cloud_cli
  role: reader

task_refs:
  contract_id: catc-20260516-120000-corpus-review
  task_id: task-corpus-review-001
  permission_grant_id: grant-corpus-review-001
  handshake_id: hsp-20260516-cloud-reader-001
  registration_id: reg-gemini-reader-001
  sandbox_id: null
  worktree_run_id: null
  review_id: review-corpus-001
  memory_gate_id: mg-liya-001
  incident_id: null

boundary:
  surface: memory_gate
  action: reject_memory_proposal
  decision: rejected
  reason_code: uncertainty_high
  risk_class: R1
  uncertainty: high

```

```

artifact_refs:
  input_hash: null
  output_hash: reader-report-hash-001
  diff_hash: null
  artifact_manifest_hash: null
  test_report_hash: null
  rollback_plan_hash: null
  upstream_event_refs: []
  downstream_event_refs: []

privacy:
  privacy_class: WP-MEMORY
  retention_class: WR-MEMORY-GATE
  raw_content_embedded: false
  secret_embedded: false
  private_memory_embedded: false
  sealed_material_embedded: false
  legal_sensitive_embedded: false
  child_data_embedded: false

review:
  reviewer_required: true
  reviewer_ref: review-corpus-001
  c_gate_required: true
  c_gate_ref: cgate-liya-002
  human_gate_required: false
  human_gate_ref: null
  legal_review_required: false
  legal_review_ref: null

integrity:
  canonicalization: yaml_canonicalization
  event_hash: null
  previous_event_hash: null
  chain_id: chain-memory-gate-liya-001
  signature_required: false
  signature_ref: null
  append_only_required: true

outcome:
  next_required_action: none
  status_after_event: completed

notes:
  summary: Agent report retained as operational reference but not promoted into
reviewed memory.
  limitations:
    - Source uncertainty remains high.
  assumptions: []

```

31.4 Red-line prohibited action attempt

```

cli_agent_witness_event:
  schema_version: cli-agent-witness-event-0.1
  event_id: we-20260516-0004
  event_family: cli_agent.anomaly.external_counteroperation
  event_class: WE-X
  timestamp: "2026-05-16T19:00:00Z"
  sequence_number: 31

```

```
governing_entity:
  entity_id: ester
  entity_name: Ester
  continuity_ref: null

agent:
  agent_id: unknown-helper-01
  agent_name: Unknown Helper
  provider: unknown
  runtime: unknown
  role: null

task_refs:
  contract_id: null
  task_id: null
  permission_grant_id: null
  handshake_id: hsp-invalid-unknown-full-access
  registration_id: null
  sandbox_id: null
  worktree_run_id: null
  review_id: null
  memory_gate_id: null
  incident_id: incident-unknown-agent-001

boundary:
  surface: anomaly
  action: prohibited_external_counteroperation_attempt
  decision: quarantined
  reason_code: external_counteroperation_attempt
  risk_class: RX
  uncertainty: medium

artifact_refs:
  input_hash: null
  output_hash: null
  diff_hash: null
  artifact_manifest_hash: null
  test_report_hash: null
  rollback_plan_hash: null
  upstream_event_refs: []
  downstream_event_refs: []

privacy:
  privacy_class: WP-INCIDENT
  retention_class: WR-INCIDENT
  raw_content_embedded: false
  secret_embedded: false
  private_memory_embedded: false
  sealed_material_embedded: false
  legal_sensitive_embedded: false
  child_data_embedded: false

review:
  reviewer_required: true
  reviewer_ref: null
  c_gate_required: true
  c_gate_ref: null
  human_gate_required: true
  human_gate_ref: null
  legal_review_required: false
  legal_review_ref: null
```

```
integrity:
  canonicalization: yaml_canonicalization
  event_hash: null
  previous_event_hash: null
  chain_id: chain-incident-unknown-agent-001
  signature_required: true
  signature_ref: null
  append_only_required: true

outcome:
  next_required_action: revoke
  status_after_event: quarantined

notes:
  summary: Prohibited live counter-operation attempt. Agent not admitted. Quarantine
and human review required.
  limitations:
    - Provider unresolved.
  assumptions: []
```

32. Validation workflow

```
parse event
-> structural validation
-> event family validation
-> privacy class validation
-> raw content prohibition check
-> reference consistency check
-> required witness check
-> chain linkage check
-> retention assignment
-> hash/canonicalization if required
-> accept / hold / quarantine
```

33. Semantic validation rules

33.1 Event family / class consistency

WE-X events must not resolve to `allowed` or `completed` without review.

33.2 Raw content flags

If any embedded sensitive-content flag is true, the event must be rejected or moved to a separate raw evidence exception route.

33.3 Required gate consistency

If `human_gate_required` is true, `human_gate_ref` should exist before final integration.

33.4 Agent self-witness rule

The agent that performed a material action may produce a report, but it cannot be sole validator of witness adequacy for that action.

33.5 Missing task contract rule

Material execution events without a task contract must be held or quarantined unless they are emergency sentinel events under pre-approved policy.

33.6 Privacy class escalation rule

If a lower privacy class event references higher privacy material, the event must be reclassified or redacted.

33.7 Chain continuity rule

If a chain uses previous event hashes, a broken hash link triggers witness anomaly review.

34. Failure mapping

Failure	Required default
malformed event	hold
event embeds secret	freeze_and_escalate
event embeds private/sealed memory	quarantine
required reference missing	hold
missing witness for privileged transition	hold / freeze
broken witness chain	quarantine / revalidate
event family unknown	hold
event class inconsistent with decision	hold
red-line event allowed as normal	revoke_and_quarantine
signature/hash mismatch	quarantine
cloud leakage suspected	freeze_and_escalate

35. Conformance levels

Level	Meaning
WEP-0	no witness event discipline
WEP-1	human-readable operational logs only
WEP-2	structured events for task and permission transitions
WEP-3	structured events + privacy/retention classes + artifact references
WEP-4	append-only witness chains + required privileged-transition witnesses
WEP-5	high assurance: canonicalized, hashed/signed where needed, challengeable, retention-governed, rollback-linked
WEP-X	non-conformant / missing witness / red-line event mishandled

36. Mandatory conformance gates

Gate	Name	Blocking failure
G0	Structured event object	only free-text logs for privileged transitions
G1	Privacy flags	no privacy class or raw-content flags
G2	Retention class	no retention policy
G3	Reference linkage	no task/permission/sandbox reference for material events
G4	Append-only correction	silent modification of witness records
G5	Required witness	privileged transition lacks witness
G6	Missing witness handling	system continues despite missing required witness
G7	Red-line event handling	prohibited action not quarantined/escalated
G8	Chain integrity	broken chain ignored
G9	Raw content exclusion	secrets/private/sealed data embedded by default
G10	Agent self-certification	agent self-certifies its own material action

37. Red-line failures

A system MUST be classified as WEP-X if:

1. privileged transitions occur without witness and without fail-closed response;
 2. witness records embed secrets by default;
 3. witness records embed private or sealed memory by default;
 4. agents self-certify their own material witness events;
 5. red-line events are treated as normal task failures;
 6. missing witness is ignored;
 7. witness chain is silently edited;
 8. prohibited external counter-operation is recorded as allowed;
 9. direct memory write attempt is not quarantined;
 10. public release is applied without witness where required;
 11. core authority mutation is recorded only in free text;
 12. witness records are used as a substitute for legal, human, or c authority.
-

38. Implementation notes

38.1 Event IDs

Recommended event ID format:

```
we-YYYYMMDD-HHMMSS-shortslug
```

38.2 Chain IDs

Recommended chain ID format:

```
chain-<surface>-<task-or-agent-or-incident-id>
```

38.3 Storage

Witness event storage SHOULD be append-friendly and reviewable.

High-assurance systems SHOULD avoid silent mutable overwrites.

38.4 Redaction

Redaction SHOULD happen before event creation where possible.

If post-event redaction is necessary, create a correction event.

38.5 UI display

Human-facing UI SHOULD display:

```
what happened
which boundary
which agent
which task
which decision
what next action
```

It SHOULD NOT expose raw sensitive material by default.

38.6 Machine-facing index

Machine-facing witness indexes SHOULD support filtering by:

- entity;
- agent;
- task;
- permission grant;
- event family;
- risk class;
- privacy class;
- retention class;
- required next action;
- unresolved anomalies.

39. Open issues

ID	Issue	Required action
0I-001	JSON Schema file	Extract normative schema to <code>.schema.json</code> .
0I-002	Canonicalization profile	Define exact JSON/YAML canonicalization.
0I-003	Signature profile	Define signing and verification policy.
0I-004	Storage backend	Define file/log/database storage profiles.

ID	Issue	Required action
0I-005	Retention policy	Define default retention duration by event class.
0I-006	UI rendering	Define human-readable display rules.
0I-007	Raw evidence sidecar	Define separate raw evidence reference object.
0I-008	Chain challenge process	Define how c, human anchor, or auditor challenges a chain.
0I-009	Cross-c witness isolation	Define isolation between Ester, Liya, and other c entities.
0I-010	Repo placement	Decide final GitHub path and package index integration.

40. Closing rule

Witness is not bureaucracy.

It is how a **c** prevents executable hands from becoming invisible authority.

Final rule:

If a worker crosses a boundary, record the boundary.
If the record is missing, do not pretend the boundary was safe.